

# PRATHAM PANCHAL

prathampanchal02994@gmail.com | prathampanchal.dev  
linkedin.com/in/pratham-panchal-abcd2994 | github.com/Pratham2994

## EDUCATION

### K. J. Somaiya College of Engineering

B.Tech in Information Technology, Honours in Cyber Security - CGPA: 9.54 (Rank 3 in branch)

2022 – 2026

Mumbai, India

## SKILLS

**Languages:** Python, C++, Java, JavaScript, TypeScript, SQL

**Frameworks & Libraries:** Node.js, React, FastAPI, Express.js, Tailwind CSS

**Tools & Technologies:** Git/GitHub, Docker, Apache Airflow, dbt, Snowflake, PostgreSQL, MongoDB, Postman

**Machine Learning & AI:** scikit-learn, LightGBM, llama.cpp, Model Context Protocol (MCP), OpenAI Whisper

## EXPERIENCE

### Barclays – Software Engineer

July 2026 – Present, Pune, India

### Colgate Global Business Services – Data Engineer Intern

Jan 2026 – June 2026, Mumbai, India

- Migrated retail sales data pipelines from legacy scripts to an ELT framework on **Snowflake** and **dbt**, scheduled with **Apache Airflow**, handling **900K+** rows a day and over **1B records** of history.
- Built incremental dbt models and SQL validation queries that matched legacy production tables before cutover.

### Barclays – Technology Intern

June 2025 – July 2025, Pune, India

- Built a **Python test automation utility** for the **Cyber Security** division that cut manual testing time for the testing team.
- Received a **Pre-Placement Offer (PPO)** for a full-time role based on performance.

## PROJECTS

### OmniCompiler – Multi-Language Debugging Platform *(research paper under review)*

2025 – 2026

- Designed a unified debugging interface for **Python, JavaScript, Java, C++ and Go** through five adapters that map each native debugger to one message protocol, with **10 of 12** debug actions supported in all five.
- Isolated every execution in a resource-limited **Docker** container that starts in about **300 ms**, and trained two ML models for language detection and breakpoint suggestion (**F1 score 0.86**). *(Python, FastAPI, Docker, React, scikit-learn)*

### FloatChat – Conversational Interface for Ocean Data *(Smart India Hackathon 2025)*

2025

- Built a natural-language assistant for **Argo float data** in which an LLM selects one of five **Model Context Protocol (MCP)** tools that execute real **PostgreSQL** queries, so every answer is grounded in data.
- Developed tools for time-series, comparison, heatmap, map and SQL outputs, with responses streamed to a chat interface together with the generated chart or map. *(Python, FastMCP, PostgreSQL, ChromaDB, Node.js, Express.js, React)*

### Local LLM Lab – Benchmark Harness for Local Language Models

2026

- Developed a benchmark harness on **llama.cpp** that executed **4,101 runs** across 41 model configurations on a 12 GB GPU, validating each answer by running unit tests or comparing SQL result rows.
- Introduced **correct answers per hour** as the ranking metric, showing that the most accurate model was not the most productive and that 5 of 6 fine-tuned models scored below their base model. *(Python, llama.cpp, CUDA)*

### Prats-Deck – Touch-Screen Control Deck on a Raspberry Pi Pico 2 W

2026

- Wrote **C++ firmware** for 15 touch apps and raised rendering from **5 to 40 frames per second** with faster drawing code and the second processor core, verified by a PC simulator and a USB test script. *(C++, Python)*

### MalShield – Malware Detection Platform

2025

- Built a malware detection pipeline for **5 file categories** (Windows and Linux executables, PDF, Office, RTF) combining a **LightGBM** classifier, **74 YARA rules** and macro extraction for static analysis, with dynamic analysis in an isolated **Docker** sandbox. *(Python, FastAPI, LightGBM, YARA, Docker)*

## ACHIEVEMENTS & LEADERSHIP

3rd place, iDEA Hackathon (Union Bank of India): bank service platform with **facial recognition** login **March 2025**

Student Placement Coordinator: ran campus placement drives with **50+ companies**

July 2024 – June 2026